

WEEK 4 STATUS REPORT

Problem 1: Multi-Site Enterprise Network

Report Date	June 9, 2026 (Monday)
Reporting Period	Week 4 — Final Testing & Documentation
GitHub Repository	https://github.com/ZohaibAhmad260/CCP-CNS2026-Problem1

1. Executive Summary

Week 4 marks the completion of all core technical deliverables for the CCP-CNS2026 Problem 1 project. The team successfully conducted comprehensive network connectivity and security testing across all three VLANs (Sales VLAN 10, Engineering VLAN 20, Admin VLAN 30). All 12 test cases passed, confirming that inter-VLAN routing, ACL-based security policies, DHCP assignment, and DoS mitigation are functioning correctly. Final GNS3 project files, running configurations, and all documentation have been prepared for GitHub submission.

2. Team Contributions — Week 4

Team Member	Role	Week 4 Contribution
Zohaib Ahmad	PM & Network Architect	Executed all ping tests, verified ACL behavior, managed GNS3 project save, coordinated deliverables
M. Efan Amir	Security Engineer	Verified ACL hit counts (BLOCK_DOS_ATTACKER: 45 hits), confirmed security policies active
Muhammad Abdullah	Automation Lead & QA	Compiled test matrix, validated pass/fail results, QA sign-off
Tayyab Hassan	Documentation Lead	Authored Week 4 status report, updated GitHub README, organized all deliverable files

3. Week 4 Completed Tasks

Status	Task	Details
✓	Ping Performance Tests	Ran 6 inter-VLAN ping tests + gateway test across PC1, PC2, PC3. All results recorded with latency values.
✓	ACL Verification	Confirmed BLOCK_SALES_TO_ADMIN and BLOCK_DOS_ATTACKER ACLs active and enforcing correctly. ICMP type:3, code:13 responses verified.
✓	Final Test Matrix	Completed 12-test-case matrix with pass/fail results, latency data, and ACL behavior documentation.
✓	Running Config Saved	R1 running configuration exported as R1_running_config.txt for all VLANs, DHCP pools, and ACLs.
✓	GNS3 Project v3 Saved	Final topology saved as CCP_Problem1_Week4.gns3 on E: drive.

✓	Week 4 Status Report	This document — covering all completed and pending deliverables.
✓	GitHub Push	All Week 4 files pushed to repository with commit message "Week4: Final testing complete, all 12 TCs passed".

4. Test Results Summary

Total Test Cases: 12 Passed: 12 Failed: 0 Pass Rate: 100%

Key Findings:

- Sales VLAN (PC1) is correctly isolated from Engineering and Admin VLANs by BLOCK_SALES_TO_ADMIN ACL.
- DoS mitigation ACL (BLOCK_DOS_ATTACKER) successfully blocks all ICMP traffic from attacker IP 192.168.10.11.
- Engineering (PC2) and Admin (PC3) VLANs can communicate bidirectionally with average latency under 70ms.
- All three DHCP pools are functioning correctly — PCs retain correct IP assignments.
- Gateway reachability confirmed for VLAN 20 (192.168.20.1, avg 28.3ms).
- SSH access to R1 remains operational (verified in Week 2, not changed in Week 4).

5. Known Limitations & Mitigations

Limitation	Reason	Mitigation / Documentation
DHCP Snooping not configured	GNS3 built-in Ethernet Switch does not support the command	Documented in report. ACLs provide equivalent protection.
Kali/Ubuntu VMs not used	C: drive storage constraints on host machine	VPCS used for DoS simulation. Attack and mitigation successfully demonstrated.
First ping packet timeout	ARP resolution delay in GNS3 environment	Expected behavior. Subsequent packets succeed. Noted in test matrix.

6. Week 5 Deliverables Plan (Tuesday)

Deliverable	Assigned To	Target
Final Report (15-20 pages)	Tayyab Hassan + Zohaib Ahmad	Tuesday AM
Video Demonstration (8-10 min, YouTube unlisted)	Zohaib Ahmad	Tuesday PM
Presentation Slides (12 min)	All team members	Tuesday PM
Incident Response Report (3-5 pages)	M. Efan Amir	Tuesday AM
Peer Evaluation Forms	All team members individually	Tuesday PM

*Week 4 Status: COMPLETE ✓ | All technical deliverables done | Ready for Week 5 final submission
CCP-CNS2026 Problem 1 | Deadline: Wednesday 8:00 AM*